Mahindra University Hyderabad

École Centrale School of Engineering

Minor II (2024 - Batch)

Program: B. Tech. Branch: All

Year: II Semester: I

Subject: Probability & Statistics (MA2103)

Date: 29/10/2025 Start Time: 10:00 AM

Time Duration: 1.5 Hours Max. Marks: 20

Instructions:

1) Each question carries 5 marks.
2) All questions are compulsory.
3) Please start each answer on a separate page and ensure that you clearly number the responses.
4) It is essential to explain each step. Correct outcomes without any description will not be evaluated.

Q 1: 5 marks

If X and Y are independent Poisson variates such that

$$\mathbb{P}(X=1)=\mathbb{P}(X=2) \text{ and } \mathbb{P}(Y=2)=\mathbb{P}(Y=3).$$

Find the variance of $(X-2Y)$.

Q 2: 5 marks

Prof. Mood implemented a grading system for the final scores in *Probability and Statistics*, assuming that student scores follow a normal distribution with mean μ and standard deviation σ. The grading policy is defined as follows:

- Grade A+: Scores greater than $\mu+2\sigma$
- Grade A: Scores in the interval $(\mu+1.5\sigma, \mu+2\sigma]$
- Grade B+: Scores in the interval $(\mu+\sigma, \mu+1.5\sigma]$

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- Grade B: Scores in the interval $(\mu + 0.5\sigma, \mu + \sigma]$
- Grade C+: Scores in the interval $(\mu, \mu + 0.5\sigma]$
- Grade C: Scores in the interval $(\mu - 0.5\sigma, \mu]$
- Grade D: Scores in the interval $(\mu - 2\sigma, \mu - 0.5\sigma]$
- Grade F (Fail): Scores less than or equal to $\mu - 2\sigma$

Based on this grading scheme, answer the following questions:

(a) Calculate the percentage of students expected to receive a grade of A.

(b) Calculate the percentage of students expected to receive a grade of F (Fail).

Given: $\Phi(1.5) = 0.9332$ and $\Phi(2) = 0.9772$, where $\Phi(z)$ denotes the cumulative distribution function of the standard normal distribution.

Q 3: **5 marks**

(a) The daily return on a stock investment has a mean (μ) of 0.2% and a standard deviation (σ) of 0.8%. The distribution of daily returns is unknown. Using Chebyshev's inequality, determine the lower bound for the probability that the return on a specific day falls between -1.4% and $+1.8\%$.

[2 marks]

(b) The moment generating function (MGF) of a discrete random variable X is given by

$$M_X(t) = \frac{1}{10}(e^t + 3e^{2t} + 4e^{3t} + 2e^{4t}).$$

Find $\mathbb{P}(X \geq 3)$.

[1 mark]

(c) The joint probability mass function of (X, Y) is given by

	$Y=0$	$Y=1$	$Y=2$
$X=-1$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$
$X=1$	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{12}$
$X=2$	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{6}$

Find $\mathbb{P}(X = -1)$ and $\mathbb{P}(Y = 2 | X = -1)$.

[2 marks]

Q 4: 5 mark

(a) The joint CDF for the vector of random variable $\mathbf{X} = (X, Y)$ is given by

$$F_{X,Y}(x, y) = \begin{cases} (1 - e^{-x})(1 - e^{-y}) & x \geq 0,\ y \geq 0 \\ 0 & \text{elsewhere.} \end{cases}$$

(i) Find the marginal CDFs of X and Y. [1 mark]

(ii) Find the probability of the event $A = \{X > 1,\ Y > 1\}$. [1.5 marks]

(b) A prisoner is trapped in a cell containing 3 doors. The first door leads to a tunnel that returns him to his cell after 2 days' travel. The second leads to a tunnel that returns him to his cell after 4 days' travel. The third door leads to freedom after 1 day of travel. If it is assumed that the prisoner will always select doors 1, 2, and 3 with respective probabilities of 0.5, 0.3, and 0.2, what is the expected number of days until the prisoner reaches freedom?

[2.5 marks]